Savitribai Phule Pune University

K.K.Wagh Art's, Commerce, Science and Comp.Science College Chandori Tal-Niphad , Dist-Nashik.

A
Project Report on
"Car Rental Portal"

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CERTIFICATE

This is to certify that Ms./Mr. Joshi Sham Shashikant Exam Seat No. 37 And Ms./Mr. Gorde Suyog Shivnath Exam Seat No. 28 have successfully and satisfactorily completed and submitted the Project titled "Car Rental Portal" in partial fulfilment of T.Y.B.Sc.(CS) Lab course CS-331-FP Semester V prescribed by Savitribai Phule Pune University during the academic year 2026-27.

Project Guide | **Head of Department (Computer Science)**

Internal Examiner | **External Examiner**

ACKNOWLEDGEMENT

We would like to special thanks to all the people who have helped us to make this project successful. We would like to take this opportunity to extend my heartful gratitude to all those people without whose support and cooperation and would not have been bring our project work to success.

We are grateful to are Project Guidance of R. P. Aher (Department of Computer Science). We are thankful to our faculty members who have helped us in project.

At last we would also like to express our sincere thanks to our other Staff. Without the moral support and encouragement of them completion of this project was rather difficult.

Also special thanks to my collegians who helped me in preparation of project

Thank you very much....

Gorde Suyog Shivnath

Joshi sham Shashikant

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Introduction :

The Car Rental Portal is a **web-based application** designed to make the process of renting cars easier, faster, and more convenient. It allows customers to browse available vehicles, check rental prices, book cars online, and view their booking and payment details through a web browser. The system reduces the need for manual booking procedures and provides a user-friendly platform where customers can manage their rental activities efficiently. Since it is web-based, users can access the portal through internet-enabled devices without installing a separate mobile application. The main aim is to save time, improve accessibility, and provide a smooth car rental experience.

The project includes features such as customer registration and login, car management, booking management, payment tracking, and booking history. An administrator can manage vehicle details, customer information, bookings, and payment records through the system. The application is developed using Java, JSP, Servlets, and PostgreSQL, while Docker supports deployment and application management. The main objective of the Car Rental Portal is to provide a reliable and efficient online rental management solution that reduces manual work, maintains organized records, and improves the overall experience for both customers and administrators.

Aim and Objectives of the System :

The main aim of the Car Rental Portal is to develop a user-friendly web-based system that allows customers to view available cars, check rental prices, book vehicles, and manage payments online. The system helps administrators manage cars, customers, bookings, and payment records efficiently. Its main objective is to reduce manual work, save time, maintain accurate records, and provide a convenient car rental experience for both customers and administrators. It also provides secure login facilities to protect user accounts and information. The system helps customers check their booking status and payment history easily. Overall, it improves the efficiency, accuracy, and convenience of car rental services.

SYNOPSIS

TITLE OF THE PROJECT :- "CAR RENTAL PORTAL"

STATEMENT OF THE PROBLEMS OR EXISTING SYSTEM

Traditional car rental services often rely on manual booking procedures and paperwork, which can lead to difficulties in checking vehicle availability, managing customer information, tracking payments, and maintaining booking records. Customers may need to visit rental offices or contact providers to obtain car details and rental prices. Managing bookings and payment information manually can also increase administrative workload and the possibility of errors. Therefore, a web-based Car Rental Portal is needed to simplify the rental process and improve the efficiency of car rental management.

AIMS AND OBJECTIVE OF THE SYSTEM

The primary aim of the Car Rental Portal is to develop a centralized and user-friendly web-based platform that simplifies car rental operations and improves the experience of customers and administrators. The system will:

- Allow customers to register, log in, and view available cars.
- Display vehicle details and rental prices.
- Enable customers to book cars online and view booking history.
- Maintain customer, vehicle, booking, and payment records.
- Allow administrators to manage cars, customers, and bookings.
- Track payment details and display transaction history.
- Reduce manual work and improve the accuracy of records.

FRONT-END

The software system uses **HTML, CSS, JavaScript, JSP, and Java Servlets** to provide an interactive and user-friendly interface for customers and administrators through a web browser.

BACK-END

The system uses **Java and Java Servlets** to process application requests and manage business logic. **PostgreSQL** is used to store and manage customer information, vehicle details, bookings, and payment records. **Docker** supports application deployment and environment management.

EXISTING SYSTEM

1. Introduction

In traditional car rental businesses, activities are often managed manually using registers, paper files, spreadsheets, and phone calls. These methods are used to maintain customer details, vehicle availability, bookings, and payment records. Manual processes are time-consuming and may lead to errors, duplicate records, and difficulties in managing rental operations efficiently.

2. Description of the Existing System

a) Customer Registration

Customers provide their personal details to rental staff, who record the information manually. This may cause data entry errors and duplicate records.

b) Vehicle Availability

Staff maintain vehicle details and availability records manually. Customers must contact the rental office to check available cars and rental prices, which can cause delays and confusion.

c) Car Booking

Customers book vehicles through phone calls or office visits. Staff check availability and record booking details manually, which may result in booking conflicts and errors.

d) Payment Management

Payments and receipts are maintained using paper records or spreadsheets. Tracking transactions, pending payments, and payment history can be difficult.

e) Record Management

Customer, vehicle, and booking records are stored in separate files or registers. Retrieving and updating information takes time and increases the possibility of errors.

f) Administrative Management

Administrators manage vehicles, customers, bookings, and payments manually, requiring additional effort and making the overall process less efficient.

3. Drawbacks Of The Existing System :

The existing manual car rental system relies heavily on paperwork, registers, spreadsheets, and phone calls to manage customer details, vehicle availability, bookings, and payment records. Although this traditional method is simple to use, it has several operational and administrative limitations that reduce efficiency and increase the possibility of errors. These drawbacks highlight the need for a computerized Car Rental Portal.

1. Description of the Existing System

The existing car rental process involves several activities handled manually by rental staff and administrators. The major limitations are described below:

a) Customer Registration

Customer details are recorded manually in registers or spreadsheets. This process is time-consuming and may result in duplicate entries, incorrect information, and difficulty retrieving customer records.

b) Vehicle Availability

Vehicle details and availability are updated manually. Customers must contact the rental office to check available cars, which may cause delays and confusion regarding vehicle status.

c) Car Booking

Bookings are managed through phone calls or office visits. Manual booking records can lead to scheduling conflicts, incorrect booking dates, and difficulties managing multiple reservations.

d) Payment Management

Payment details and receipts are maintained manually. Tracking completed payments, pending amounts, and transaction history becomes difficult and may lead to errors in financial records.

e) Record Maintenance

Customer, vehicle, and booking information is stored in separate files or registers. Searching, updating, and maintaining these records requires additional time and effort.

2. Major Drawbacks of the Existing System

Manual Data Entry: Recording information manually can lead to errors, duplicate entries, and inconsistent records.

Time-Consuming Process: Managing customer details, checking vehicle availability, confirming bookings, and maintaining payment records requires considerable time.

Booking Conflicts: Lack of centralized, updated availability information can result in overlapping reservations or incorrect booking details.

Poor Communication: Customers may experience delays in receiving information about vehicle availability, booking confirmation, or payment status.

Difficult Record Tracking: Retrieving previous bookings, customer details, and payment history can be challenging.

Lack of Real-Time Updates: Customers and administrators may not have immediate access to the latest vehicle availability and booking information.

Data Security Risks: Paper records and spreadsheets may be lost, damaged, or accessed by unauthorized persons if proper safeguards are not maintained.

Limited Scalability: Managing a growing number of customers, vehicles, and reservations becomes increasingly difficult with manual methods.

Therefore, a computerized Car Rental Portal is required to reduce manual work, improve record accuracy, simplify booking management, and make car rental operations more efficient.

PROPOSED SYSTEM

1. Introduction:

The proposed Car Rental Portal is a web-based application designed to simplify and automate car rental operations. It provides a centralized platform for managing customers, vehicles, bookings, and payment details. The system reduces paperwork, saves time, improves data accuracy, and makes the rental process more convenient for customers and administrators.

2. Description of the Proposed System:

The proposed system provides the following features:

a) Customer Registration and Login: Customers can register and log in to access the portal and manage their rental activities.

b) Vehicle Management: Customers can view available cars, vehicle details, and rental prices. Administrators can add, update, or remove vehicle information.

c) Online Car Booking: Customers can select an available vehicle and submit booking details. The system maintains booking records and helps administrators manage reservations.

d) Payment Management: The system maintains payment details, payment status, and transaction records for easier tracking.

e) Booking History: Customers can view their current and previous booking details, while administrators can access booking information when required.

f) Administrative Management: Administrators can manage customer records, vehicle details, bookings, and payments through a centralized system.

g) Vehicle Availability Checking: The system helps customers identify available vehicles before making a booking, reducing confusion and booking conflicts.

h) Centralized Database: Customer, vehicle, booking, and payment information is stored in a PostgreSQL database for organized record management.

i) Booking Status Tracking: Booking information can be maintained and checked to help customers and administrators follow reservation progress.

j) User-Friendly Interface: The web interface allows users to navigate the portal and access relevant information easily.

3. Advantages of the Proposed System

- Reduces paperwork and manual effort.
- Provides easy access to vehicle and booking information.
- Improves data accuracy and record maintenance.
- Simplifies booking and payment management.
- Helps reduce vehicle reservation conflicts.
- Provides organized storage of customer and rental records.
- Saves time for customers and administrators.
- Makes it easier to update vehicle and booking information.

SCOPE OF THE SYSTEM

1. Introduction

The scope of the Car Rental Portal is to develop a web-based application that simplifies and improves the management of car rental activities. The system provides a centralized platform for handling customer information, vehicle details, bookings, and payment records. It aims to reduce manual work, improve data organization, and provide convenient access to rental services for customers and administrators.

2. Scope of the Proposed System:-

The proposed Car Rental Portal covers the following areas:

a) Customer Management:- The system allows customers to register, log in, and manage their rental activities. Customer information can be stored and retrieved efficiently when required.

b) Vehicle Management:- The system maintains vehicle details such as car name, model, registration information, rental price, and availability. Administrators can manage vehicle records according to their requirements.

c) Booking Management:- Customers can view available vehicles and submit booking requests. The system maintains booking details, including customer information, selected vehicle, rental dates, and booking status.

d) Payment Management:- The system stores payment-related information, including payment amounts and payment status. This helps administrators maintain organized payment records and track transactions.

e) Booking History:- Customers can view their booking information and previous rental records. Administrators can access booking details to support rental operations and record verification.

f) Administrative Functions:- Administrators can manage customer details, vehicle information, bookings, and payment records through a centralized interface.

g) Database Management:- The system uses a PostgreSQL database to store and organize customer, vehicle, booking, and payment information. This supports efficient data retrieval and maintenance.

h) User-Friendly Interface:- The portal provides a web-based interface that allows customers and administrators to access the system through a web browser.

3. Benefits and Limitations of the Scope

The system reduces paperwork, improves record accuracy, simplifies booking management, and saves time for customers and administrators. It also makes vehicle and booking information easier to maintain and retrieve.

The system's scope is limited to the features implemented in the project. Advanced features such as GPS-based vehicle tracking, mobile applications, automated notifications, and integration with external payment gateways may be considered for future development if they are not included in the current version.

FEATURES

The Car Rental Portal provides an easy and efficient platform for customers to rent cars online. It includes customer-side features and system-side (administrator) features for managing the rental process.

1. User Side

User Registration: Allows new customers to create personal accounts by providing the required details.

User Login: Allows registered customers to securely access their personal accounts.

View Available Cars: Displays available vehicles with their details, features, and rental prices.

Car Details: Allows customers to view complete vehicle information before making bookings.

Car Booking: Enables customers to select a vehicle and submit their rental booking details.

Payment Details: Allows customers to view payment information and transaction status.

Booking History: Enables customers to check their current and previous rental bookings.

Profile Management: Allows customers to view and update their personal account information.

2. System Side

Admin Login: Provides authorized administrators secure access to system management functions.

Customer Management: Allows administrators to view, update, and manage customer records.

Vehicle Management: Enables administrators to add, update, and remove vehicle information.

Booking Management: Allows administrators to view, track, and manage customer bookings.

Availability Management: Helps maintain accurate and updated vehicle availability records.

Payment Management: Maintains payment records, transaction details, and payment status.

Database Management: Stores and organizes customer, vehicle, booking, and payment data securely using PostgreSQL.

Record Maintenance: Supports easy retrieval, modification, and updating of stored information.

System Security: Restricts administrative functions to authorized users and protects system access.

Report and Information Access: Helps administrators review customer, vehicle, booking, and payment information efficiently.

REQUIREMENT SPECIFICATION

1. Hardware Requirements

The hardware requirements necessary to run the Car Rental Portal are:

- **Processor:** Intel Core i3 or equivalent.
- **RAM:** Minimum 4 GB.
- **Hard Disk:** Minimum 10 GB of free space.
- **System Type:** Desktop or laptop computer.
- **Internet Connection:** Required for network access where applicable.

2. Software Requirements

The software required to develop and run the Car Rental Portal includes:

- **Operating System:** Windows or Linux.
- **Front-End:** HTML, CSS, JavaScript, and JSP.
- **Back-End:** Java and Java Servlets.
- **Database:** PostgreSQL.
- **Web Server:** Apache Tomcat.
- **Containerization:** Docker and Docker Compose.
- **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox.
- **Development Environment:** Visual Studio Code or another Java-compatible IDE.

3. Functional Requirements

The functional requirements describe the operations performed by the system:

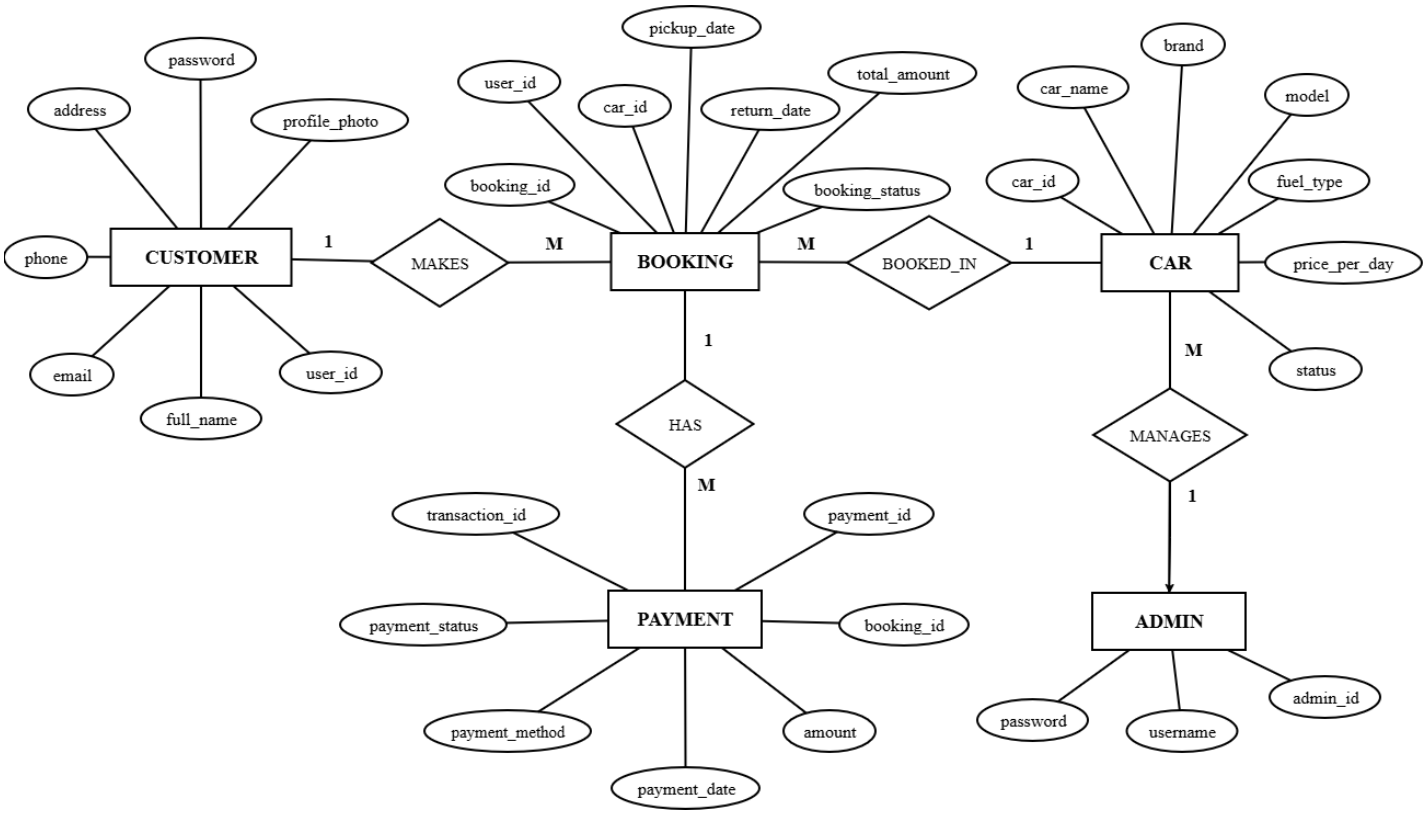
- **User Registration:** The system allows customers to create accounts.
- **User Authentication:** The system verifies login credentials.
- **Vehicle Management:** Administrators can manage vehicle details.
- **Car Booking:** Customers can submit vehicle booking requests.
- **Payment Management:** The system maintains payment details and status.
- **Booking History:** Customers can view their booking records.
- **Admin Management:** Administrators can manage customers, vehicles, and bookings.

4. Non-Functional Requirements

- **Usability:** The system should provide a simple and user-friendly interface.
- **Performance:** Pages and database operations should respond efficiently.
- **Security:** User information and administrative functions should be protected.
- **Reliability:** The system should maintain consistent and accurate records.
- **Maintainability:** The application should be easy to update and maintain.
- **Compatibility:** The portal should work with commonly used web browsers.

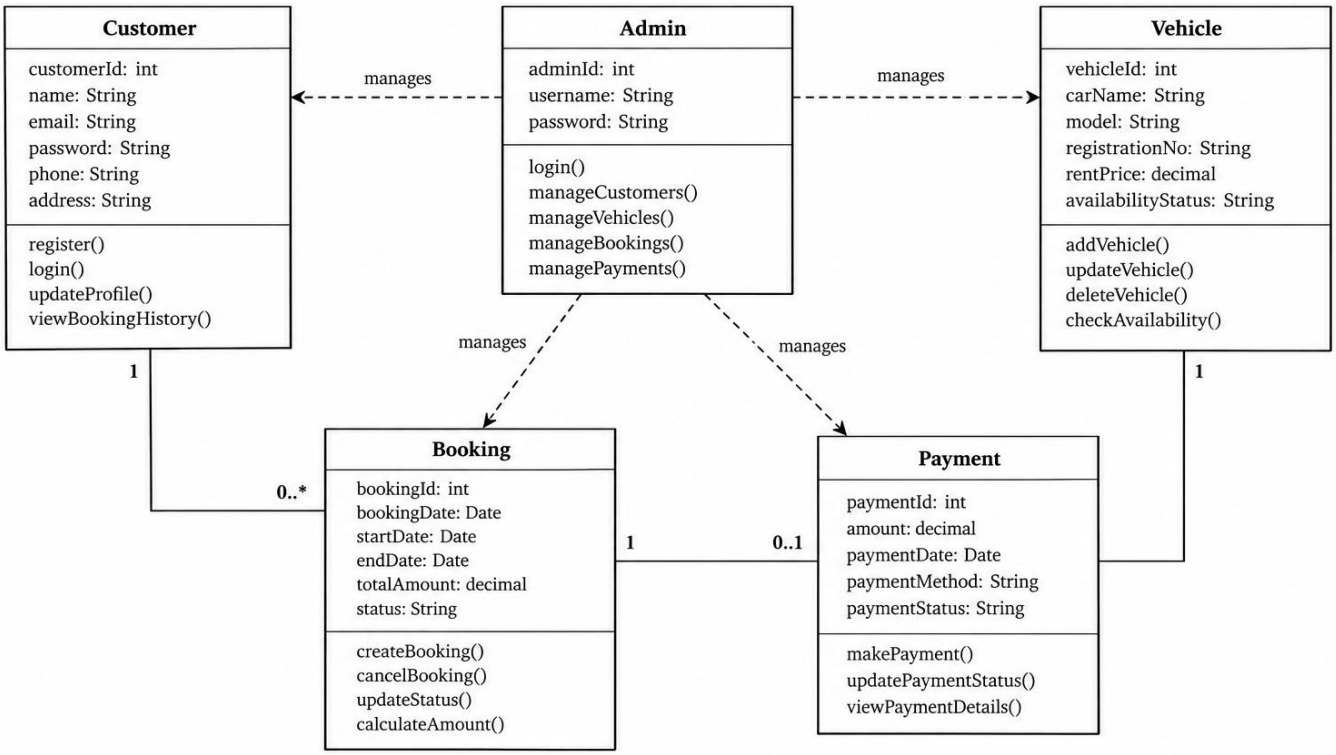
SYSTEM DESIGN

E-R DIAGRAM :-

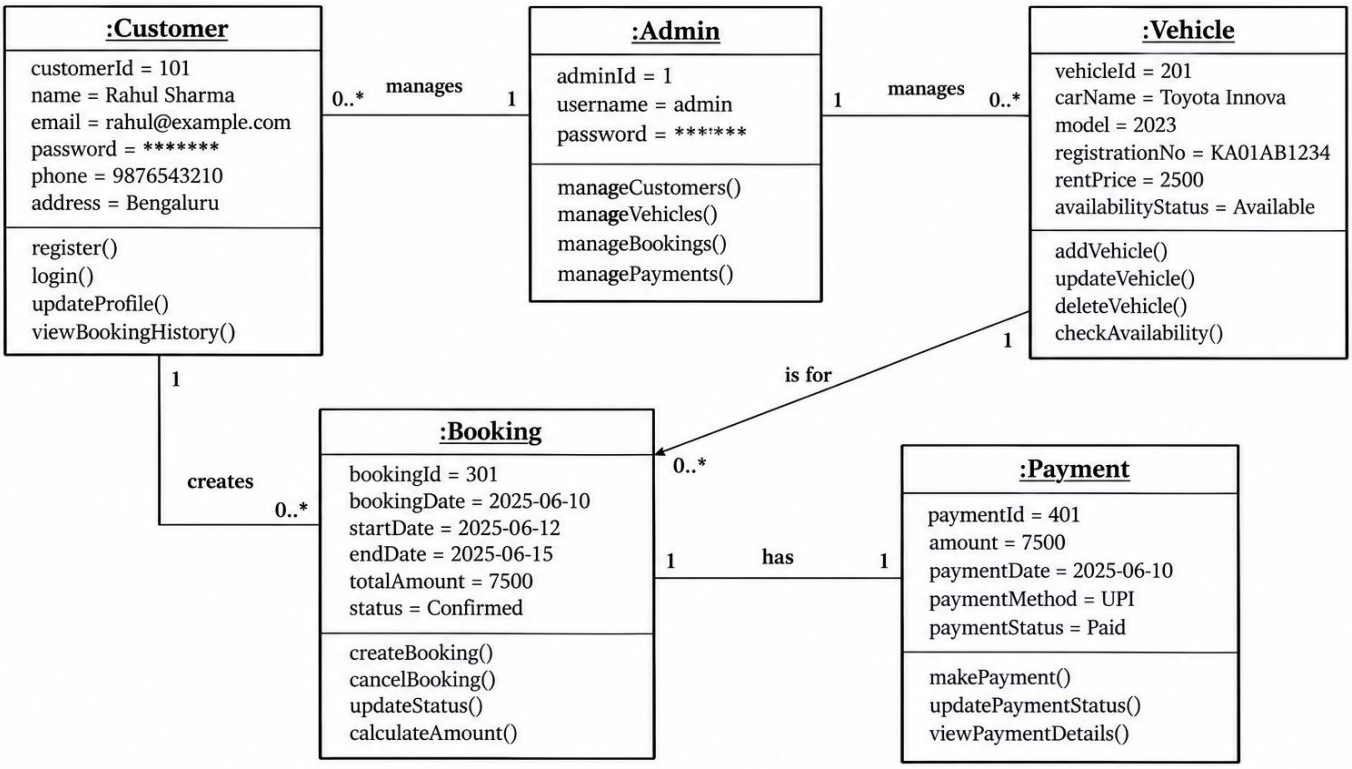


UML DIAGRAMS :

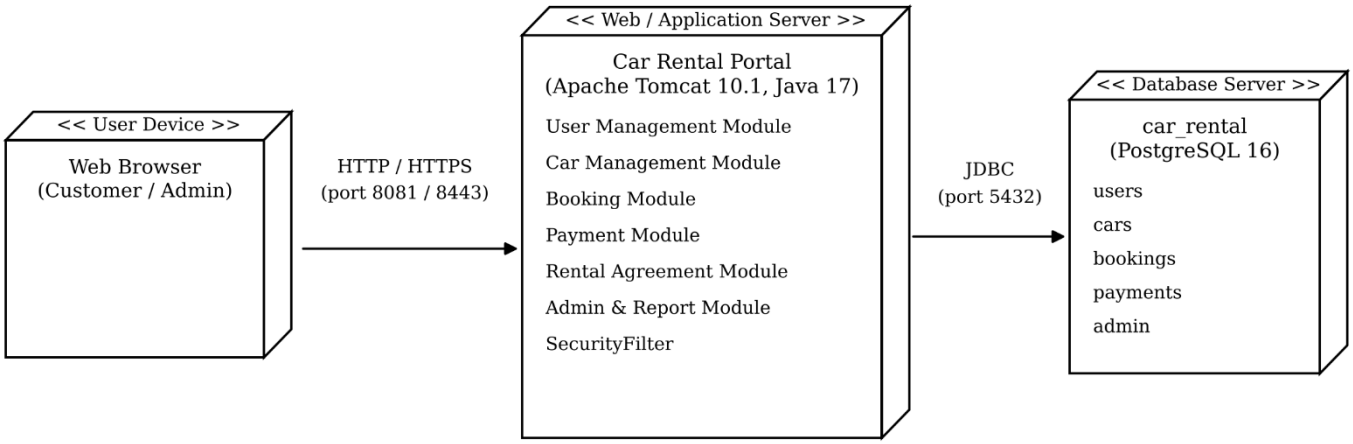
Class diagram:-



Object Diagram :-

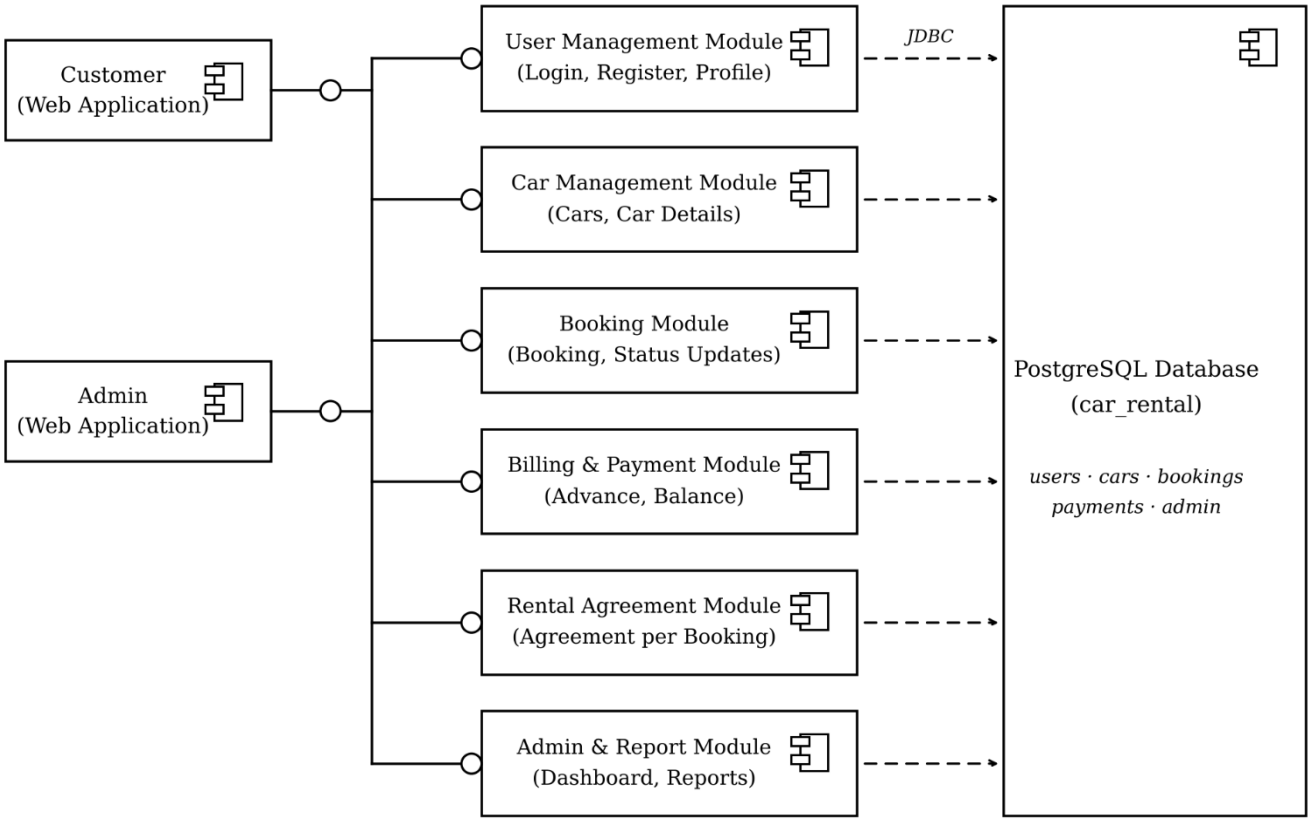


Deployment Diagram:-

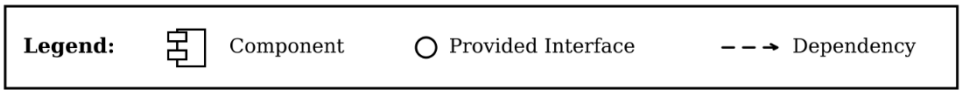


Both servers run as Docker containers (docker-compose): Tomcat serves CarRentalPortal.war, PostgreSQL holds the database.

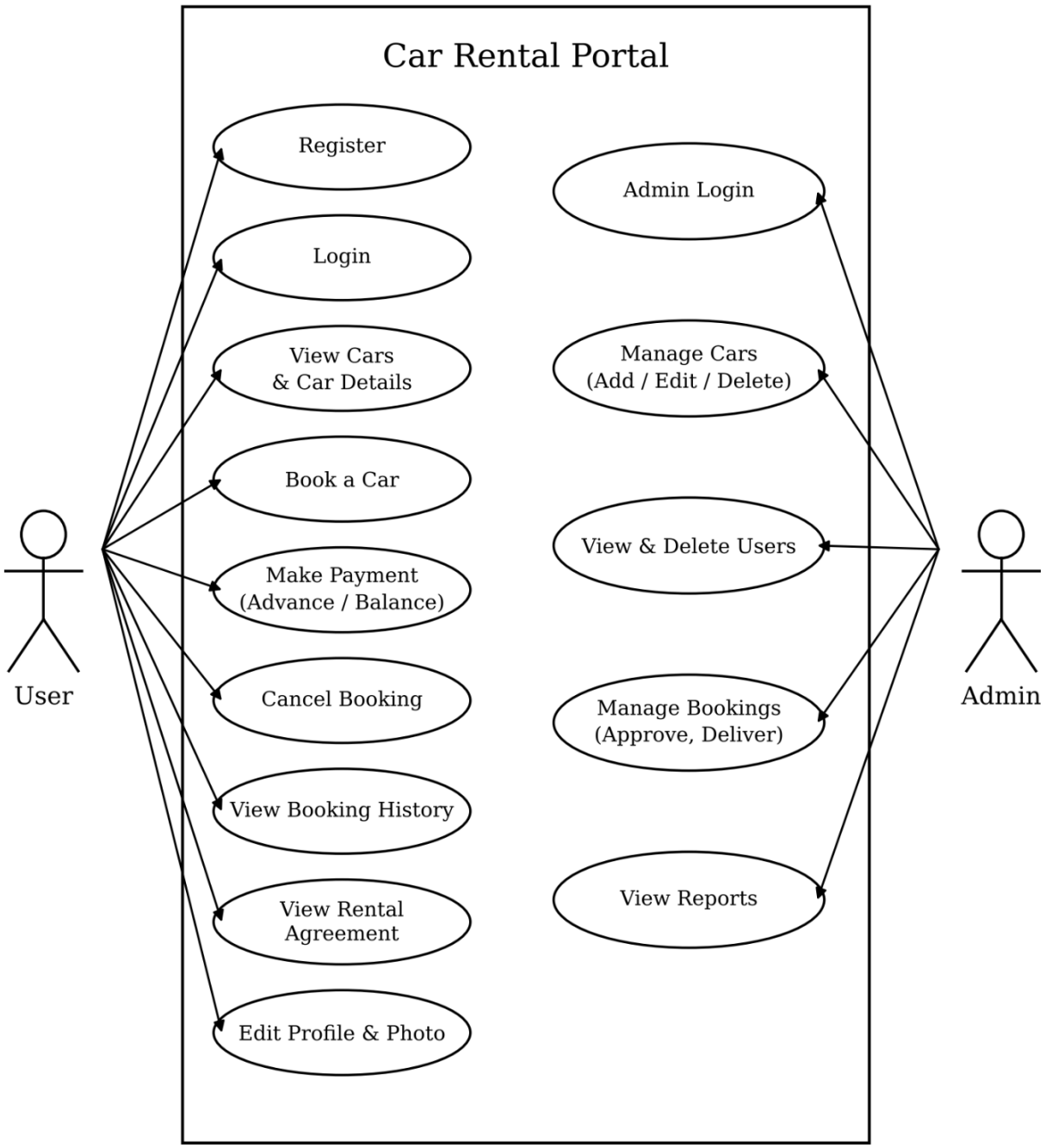
Component Diagram:-



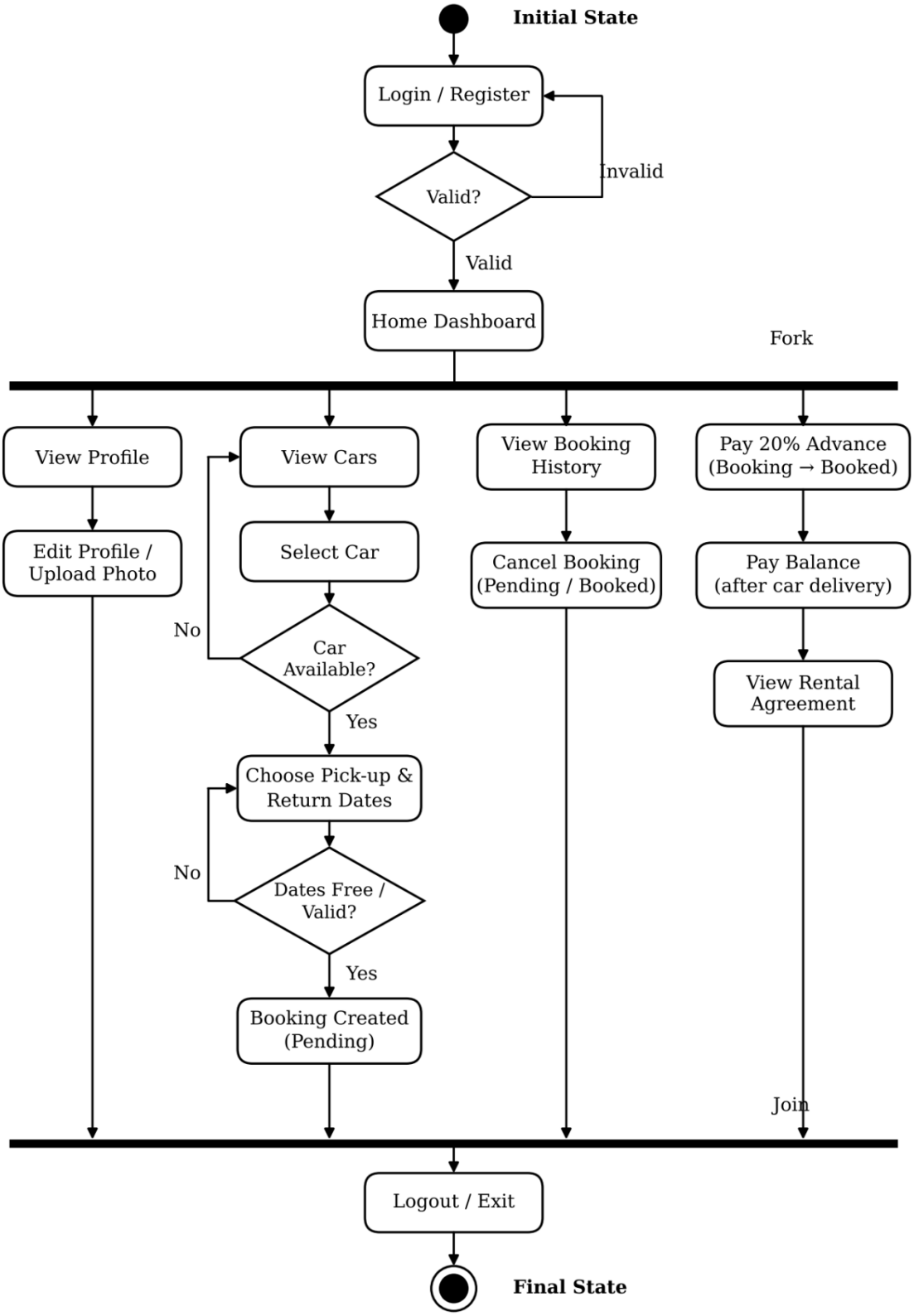
All requests pass through SecurityFilter (admin pages need admin login, CSRF token on admin forms).



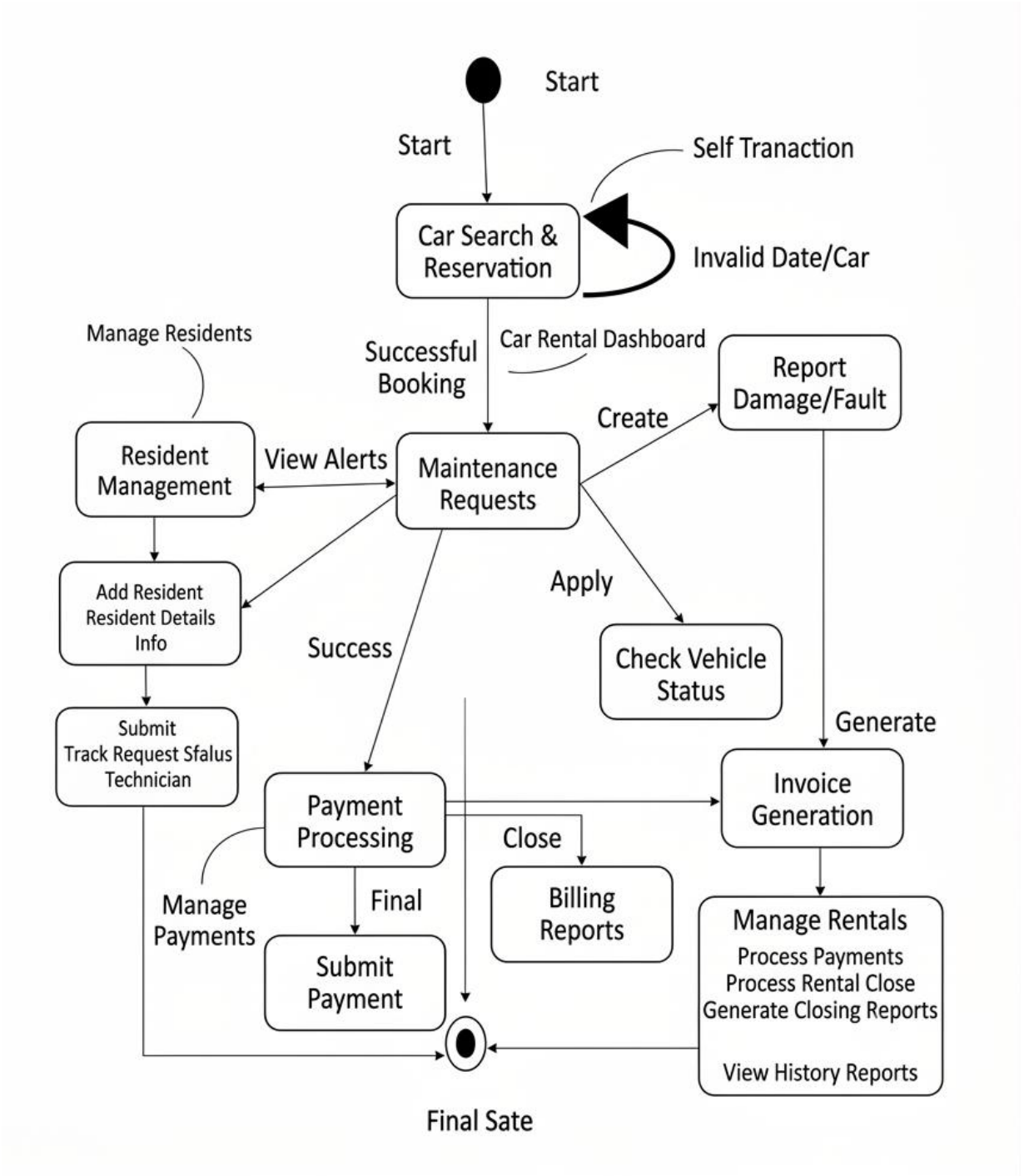
Use Case Diagram:-



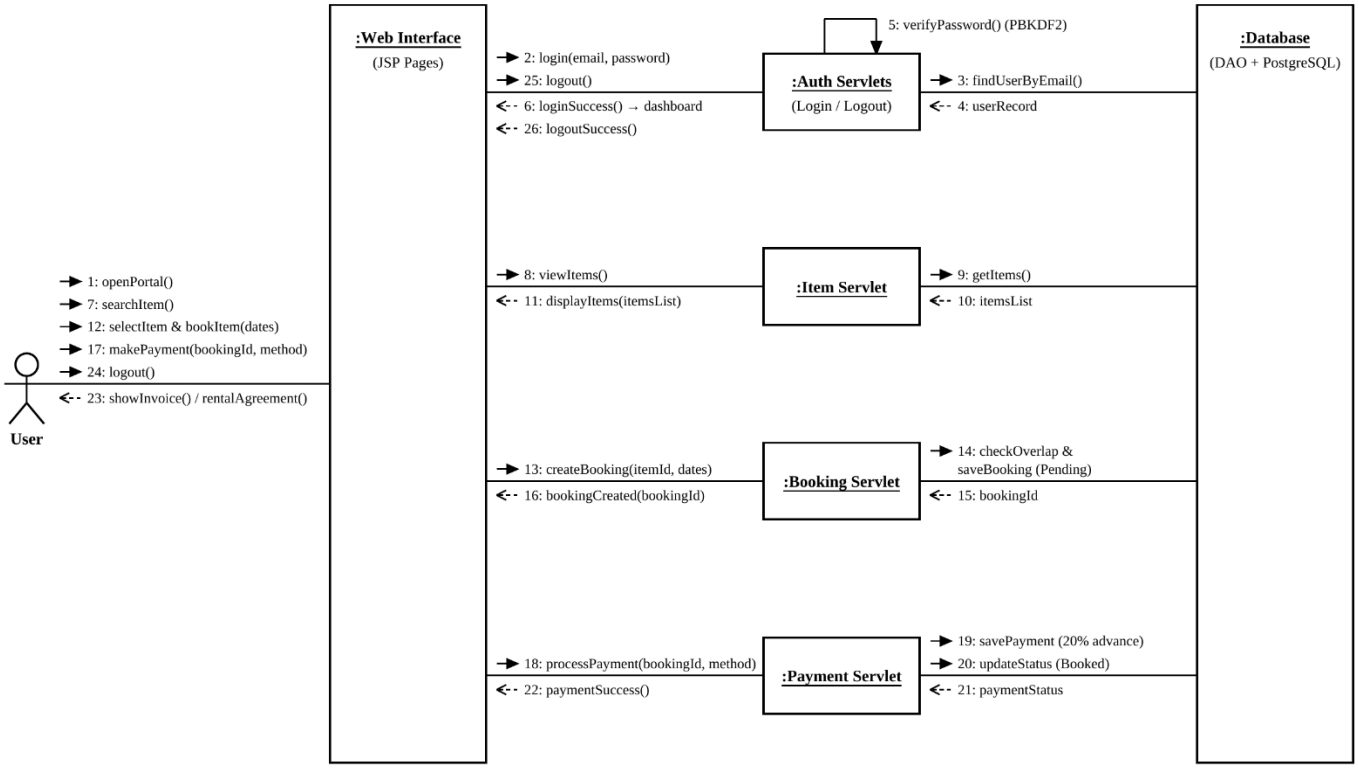
Activity Diagram:-



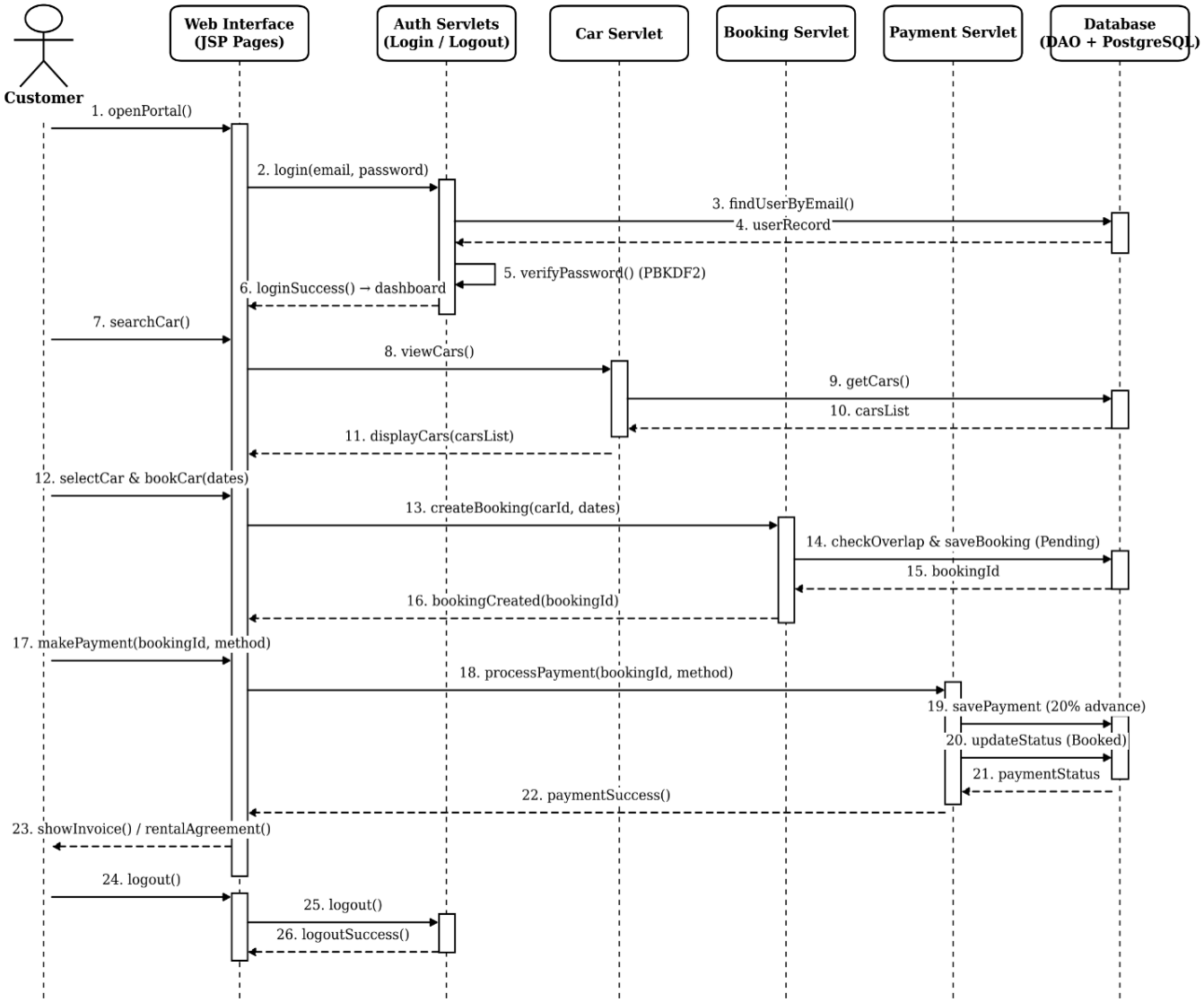
State Chart Diagram:-



Collaboration Diagram:-



Sequence Diagram:-



DATA DICTIONARY

1. Customer Table

Field Name	Data Type	Description
customer_id	INT	Primary key
name	VARCHAR(100)	Customer name
email	VARCHAR(100)	Customer email
phone	VARCHAR(15)	Contact number
password	VARCHAR(255)	Encrypted password
address	TEXT	Customer address

2. Vehicle Table

Field Name	Data Type	Description
vehicle_id	INT	Primary key
vehicle_name	VARCHAR(100)	Vehicle name
brand	VARCHAR(50)	Vehicle brand
model	VARCHAR(50)	Vehicle model
registration_number	VARCHAR(20)	Vehicle registration number
fuel_type	VARCHAR(20)	Fuel type
price_per_day	DECIMAL(10,2)	Daily rental price
availability_status	VARCHAR(20)	Vehicle availability

3. Booking Table

Field Name	Data Type	Description
booking_id	INT	Primary key
customer_id	INT	Foreign key referencing Customer
vehicle_id	INT	Foreign key referencing Vehicle
start_date	DATE	Rental start date
end_date	DATE	Rental end date
total_amount	DECIMAL(10,2)	Total rental cost
booking_status	VARCHAR(20)	Booking status
created_at	TIMESTAMP	Booking creation date

4. Payment Table

Field Name	Data Type	Description
payment_id	INT	Primary key
booking_id	INT	Foreign key referencing Booking
amount	DECIMAL(10,2)	Payment amount
payment_method	VARCHAR(30)	Payment method
payment_status	VARCHAR(20)	Payment status
transaction_id	VARCHAR(100)	Transaction reference
payment_date	TIMESTAMP	Payment date

5. Admin Table

Field Name	Data Type	Description
admin_id	INT	Primary key
username	VARCHAR(50)	Admin username
email	VARCHAR(100)	Admin email
password	VARCHAR(255)	Encrypted password

TESTING PHASE

1. Introduction

Testing is an important phase of the Car Rental Portal project. It is performed to ensure that all modules work correctly and meet the project requirements. The system is tested for functionality, database connectivity, security, usability, and error handling.

2. Types of Testing

1. Unit Testing

Individual modules, such as user registration, login, car management, and booking, are tested separately.

2. Integration Testing

The integration between JSP pages, Servlets, Java classes, and the PostgreSQL database is tested to ensure proper communication.

3. Functional Testing

All major features are tested to verify that they produce the expected results.

4. Database Testing

Database operations such as inserting, updating, retrieving, and deleting records are checked.

5. Security Testing

Password hashing, session management, input validation, and access control are tested to protect user information.

6. User Interface Testing

The website is tested on desktop and mobile screens to check navigation, layout, forms, buttons, and readability.

3. Test Cases

Test ID	Test Case	Expected Result	Status
TC01	User registration with valid details	Account created successfully	Pass
TC02	Registration with an existing email	Duplicate registration rejected	Pass
TC03	Login with valid credentials	User redirected to dashboard	Pass
TC04	Login with incorrect credentials	Error message displayed	Pass
TC05	View available cars	Available cars displayed	Pass
TC06	Book an available car	Booking details saved	Pass
TC07	Submit payment details	Payment record processed or saved as designed	Pass
TC08	View booking history	User bookings displayed	Pass
TC09	Admin login	Admin dashboard opened	Pass
TC10	Access a protected page without login	Access denied or redirected to login	Pass
TC11	Database connectivity	Application communicates with PostgreSQL	Pass
TC12	Responsive layout	Pages adjust to mobile and desktop screens	Pass

4. Testing Environment

- Operating System: Windows
- Programming Language: Java 17
- Frontend: HTML, CSS, JSP
- Backend: Java Servlets
- Database: PostgreSQL
- Build Tool: Apache Maven
- Server: Apache Tomcat 10.1
- Deployment: Docker (if used for the tested build)

5. Testing Result

The testing phase verifies whether the Car Rental Portal operates according to its requirements. The main objective is to ensure reliable user registration, authentication, car browsing, booking management, payment record handling, and administrative operations.

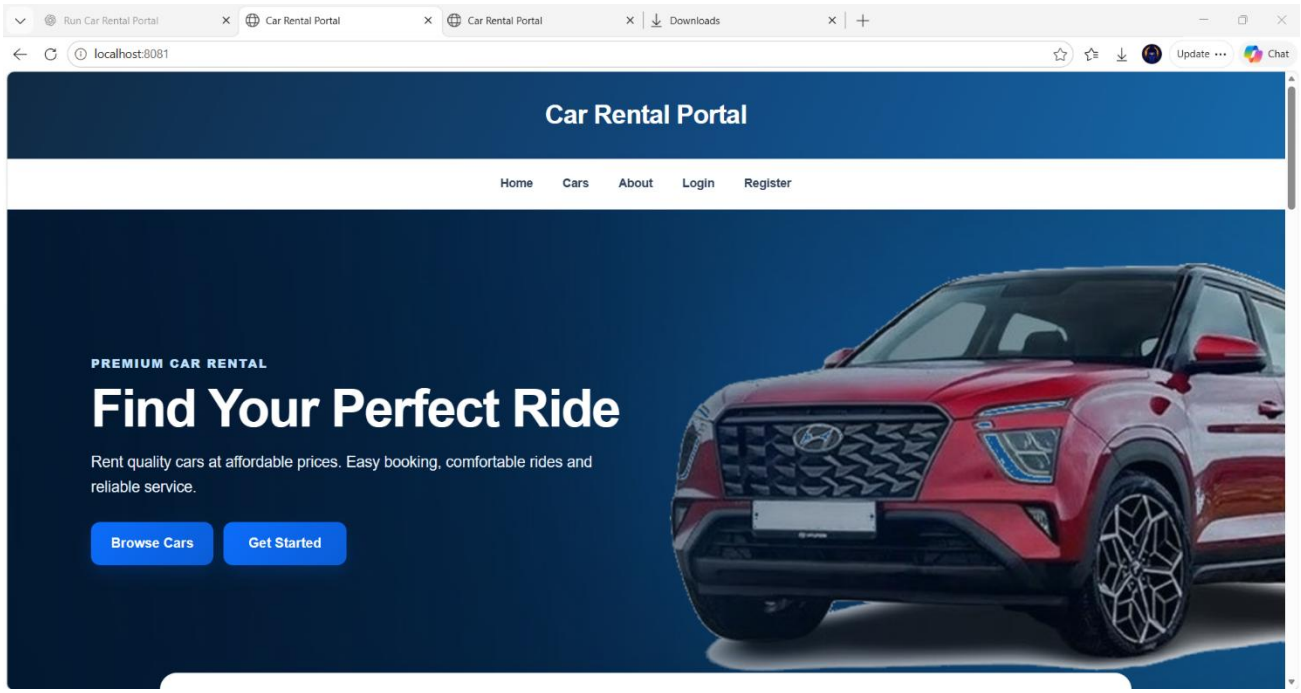
Any errors identified during testing should be corrected and the affected test cases should be executed again.

6. Conclusion

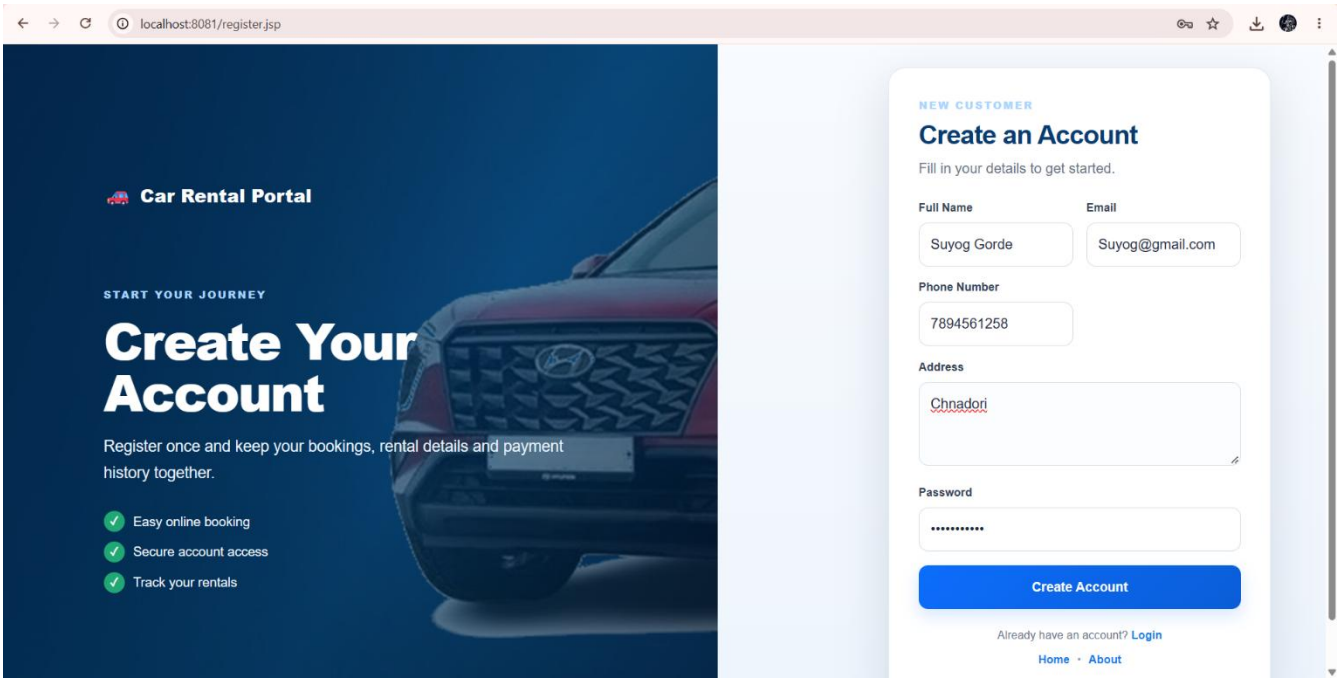
Testing helps improve the reliability, security, and usability of the Car Rental Portal. Successful completion of the planned test cases indicates that the tested modules are functioning as expected in the test environment.

SCREEN

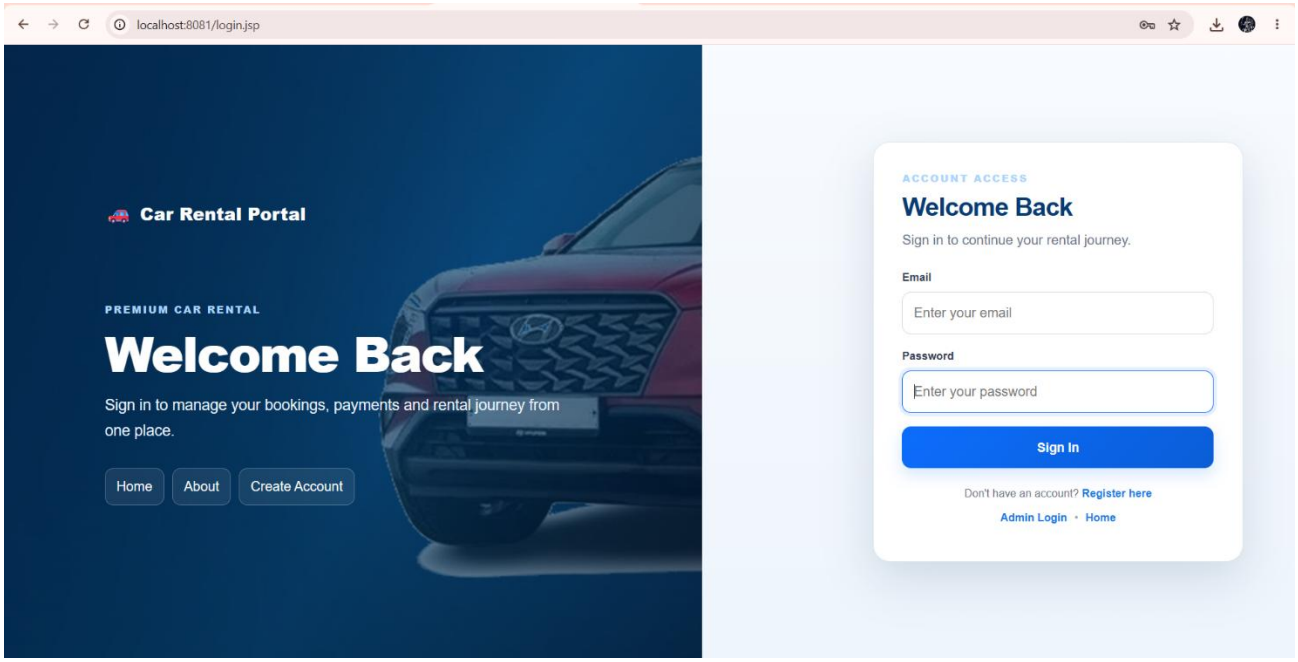
Home page:-



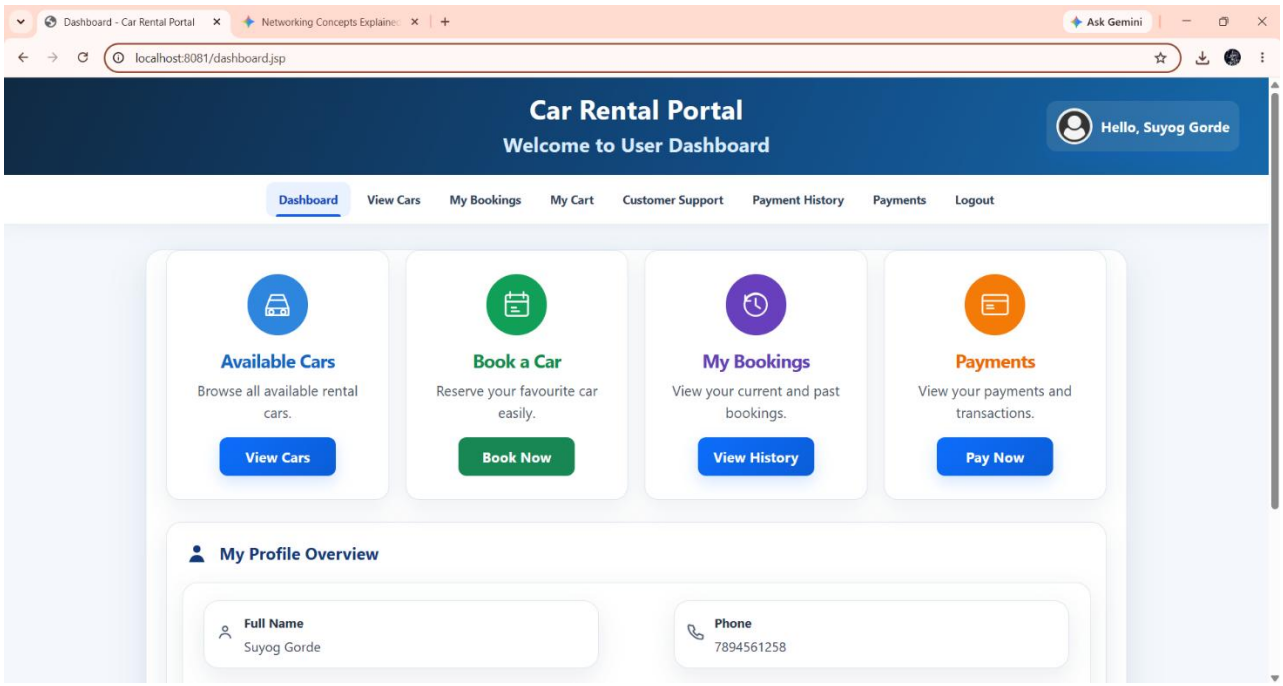
User Registration:-



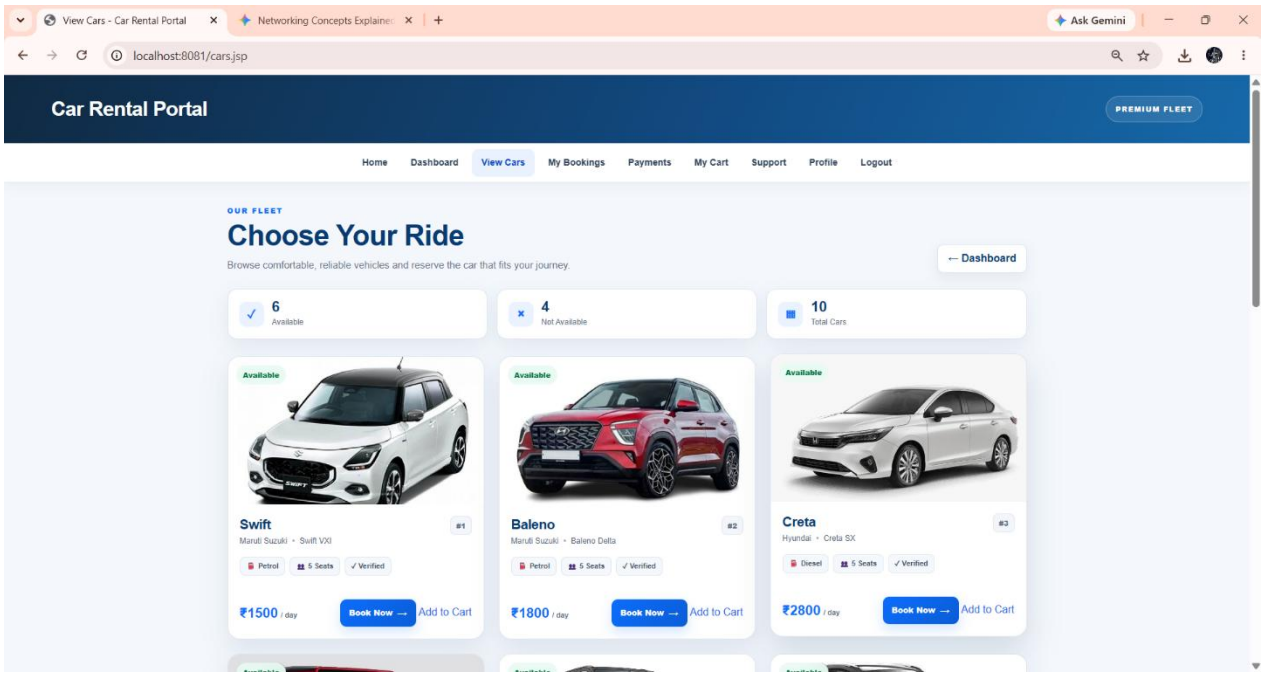
User Login:-



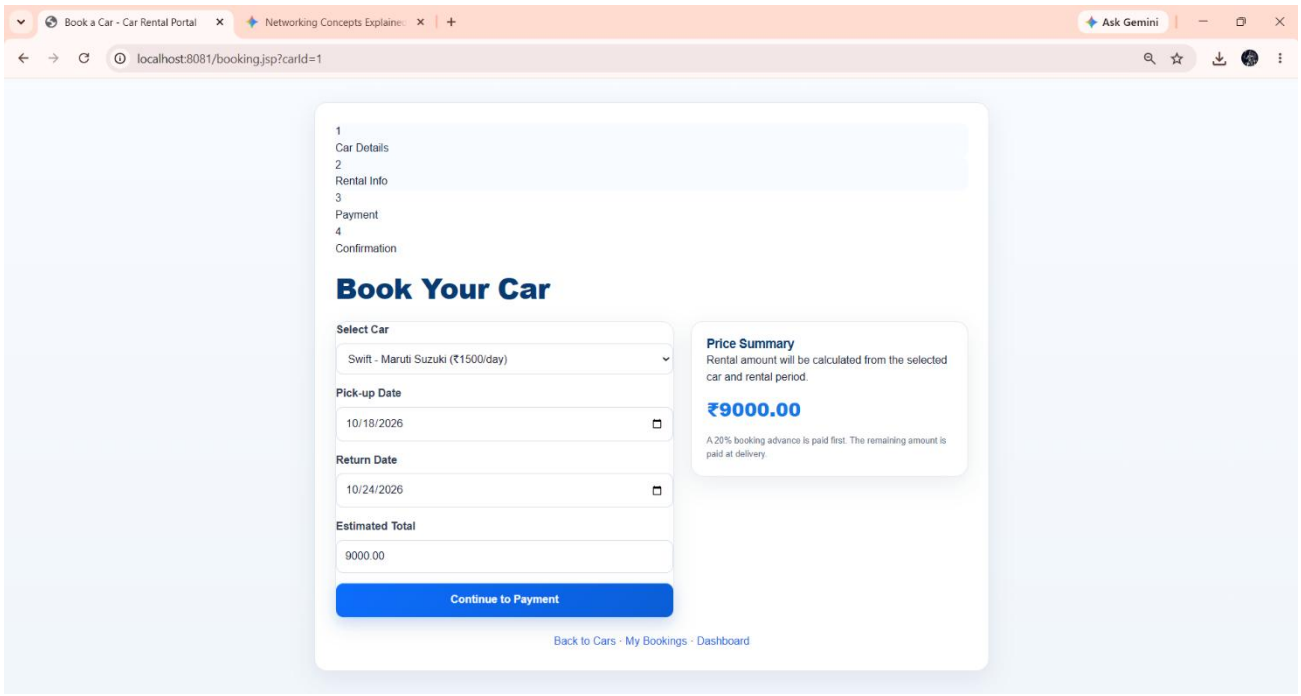
User Dashboard :-



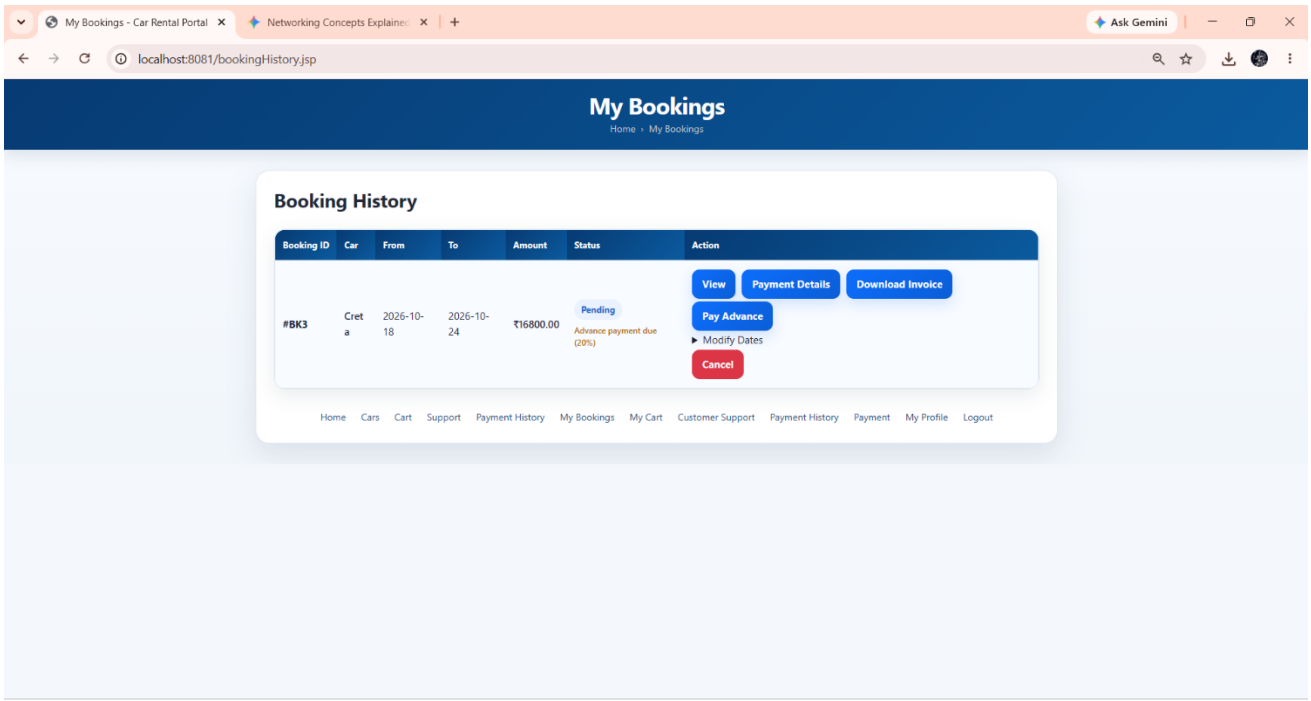
View Car :-



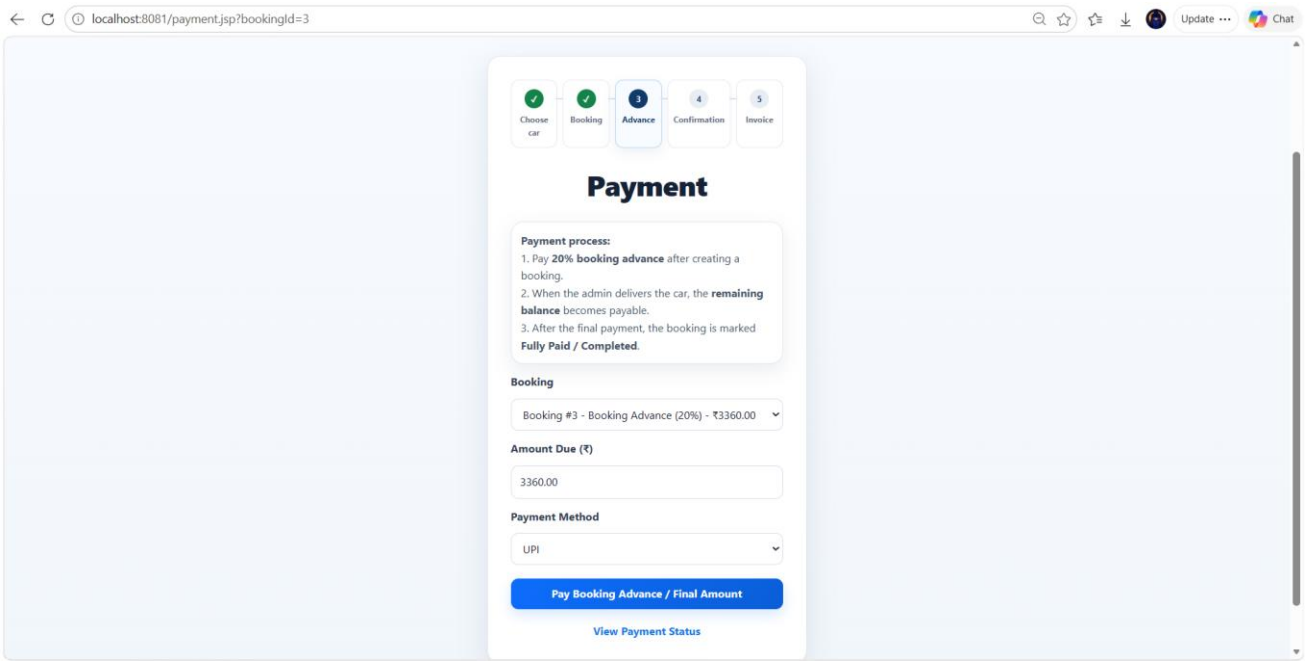
Car Booking :-



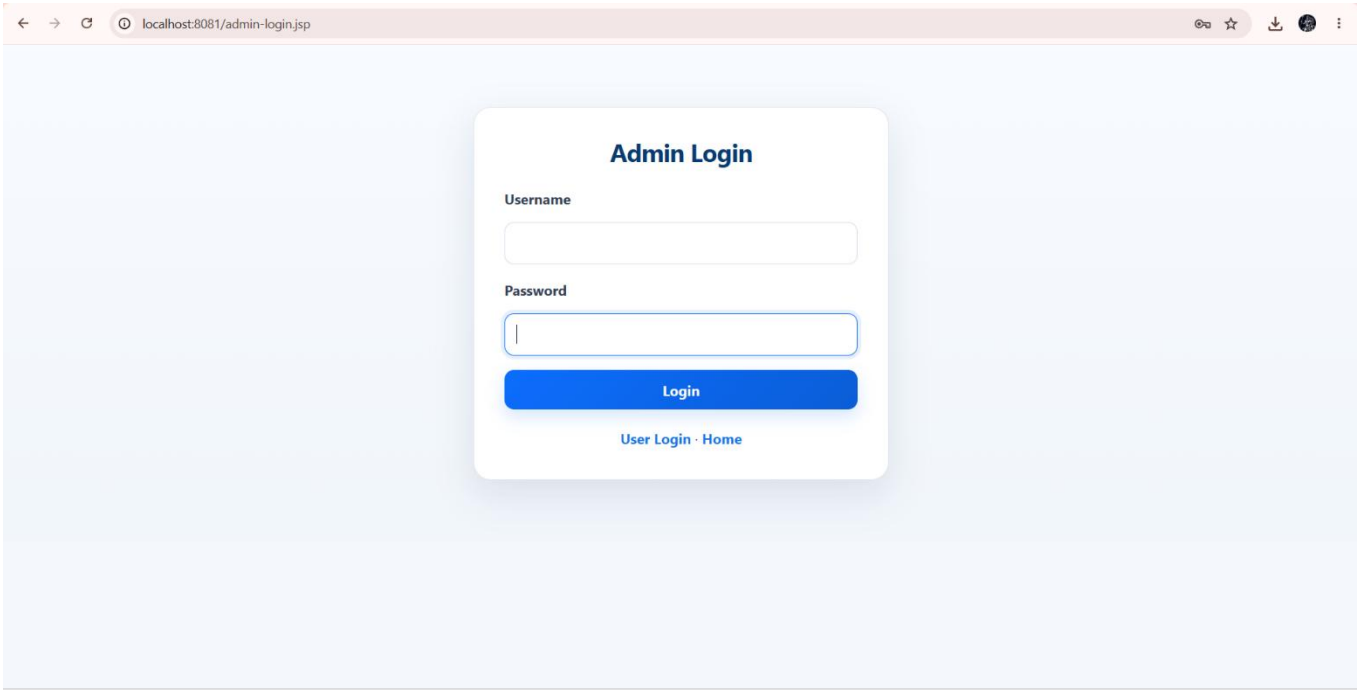
My booking:-



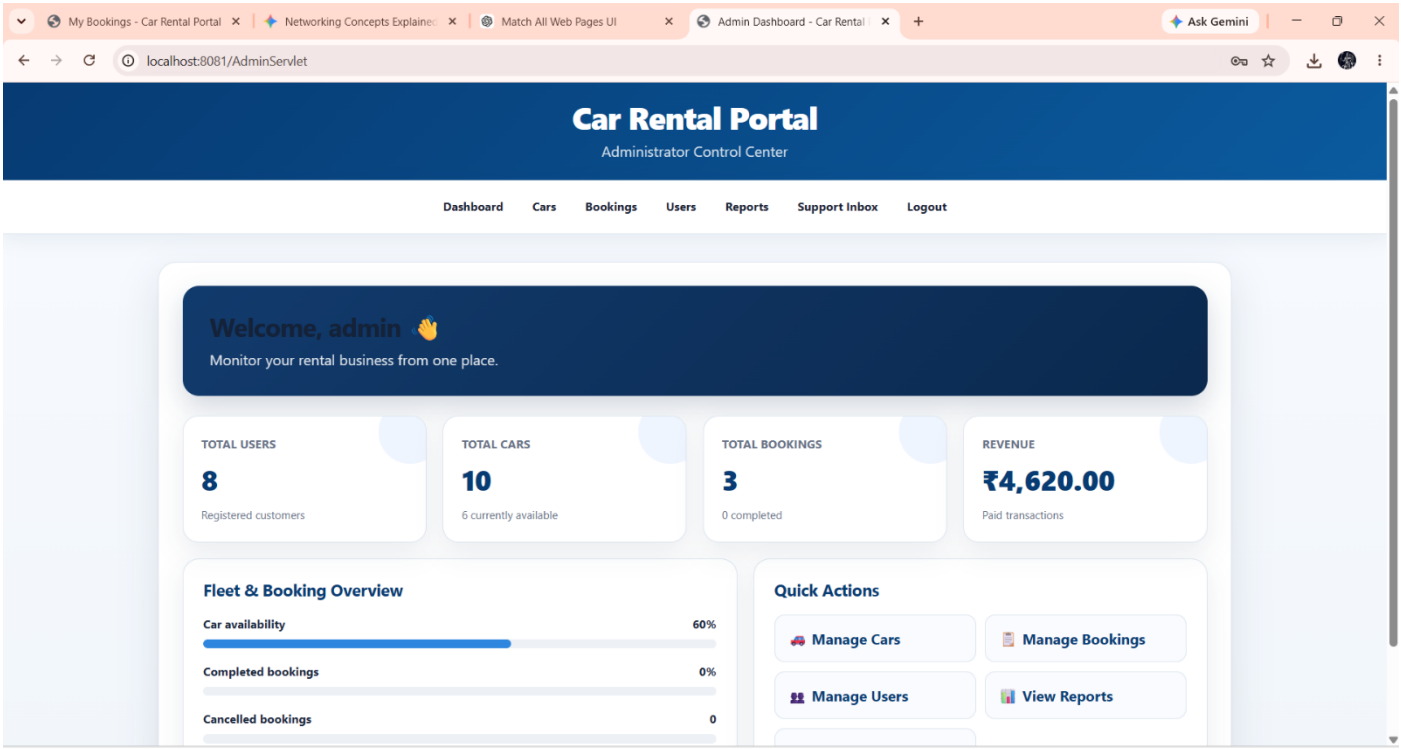
Payment:-



Admin Login :-



Admin Dashbord:-



Conclusion

The Car Rental Portal is a web-based application developed to simplify and automate the car rental process. It provides a user-friendly interface that allows customers to register, log in, view available cars, make bookings, and check their booking and payment details. The system helps reduce manual work and makes the car rental process more convenient and efficient.

The project is developed using Java, JSP, Servlets, HTML, CSS, and PostgreSQL. Apache Tomcat is used as the web server, while Maven helps manage the project build. The application includes important modules such as user authentication, car management, booking management, payment records, and an administrative dashboard. Database integration helps maintain and retrieve information systematically, while security measures help protect user data.

In conclusion, the Car Rental Portal provides a practical solution for managing car rental activities through a centralized web application. It improves accessibility, organization, and ease of use for both customers and administrators. In the future, the project can be enhanced by integrating an online payment gateway, GPS-based vehicle tracking, automated booking notifications, and advanced reporting features to make the system more reliable and feature-rich.

Bibliography

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These resources helped in understanding programming concepts, web application development, database management, server configuration, and application deployment.


Thank
you!